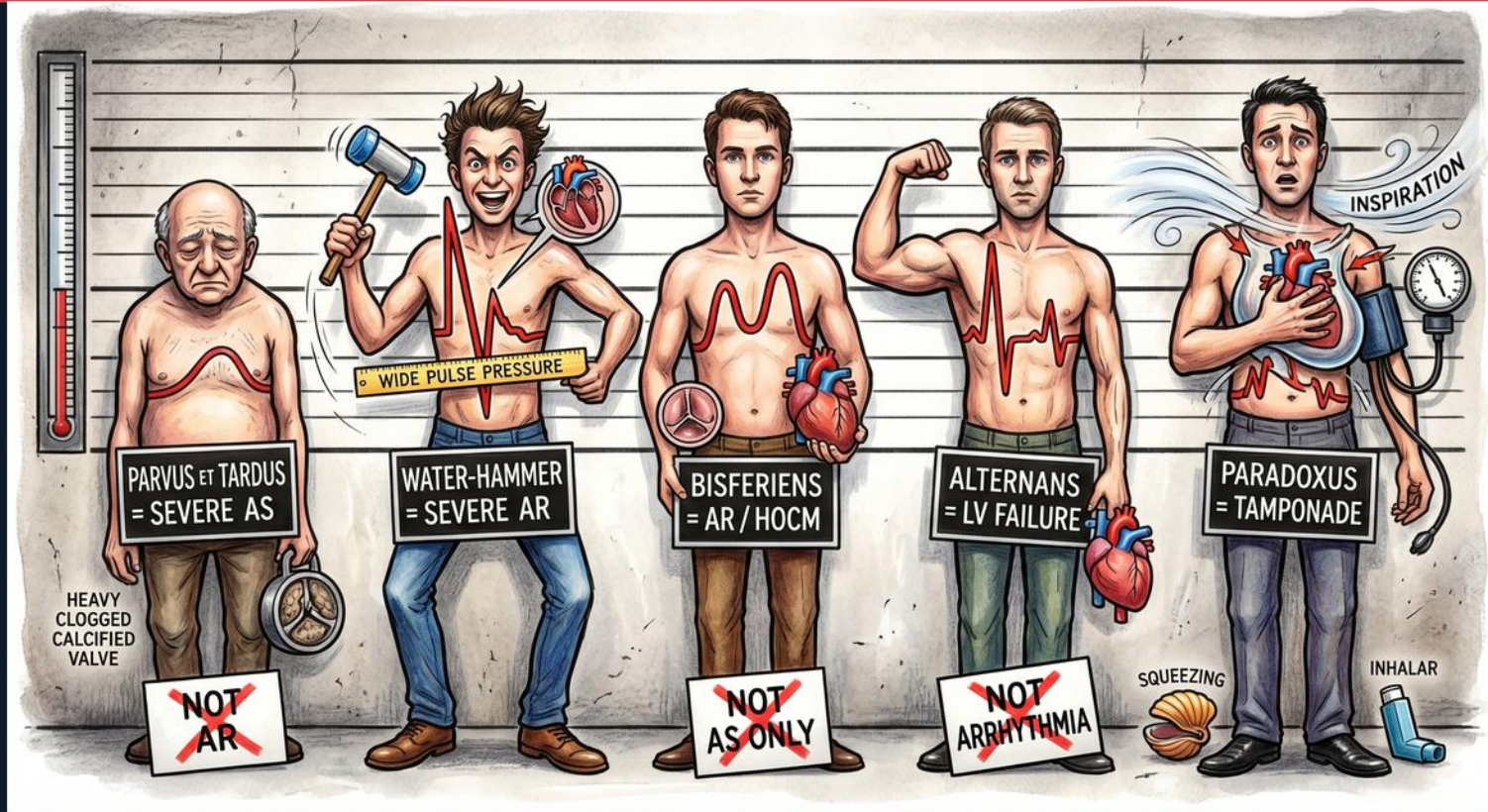
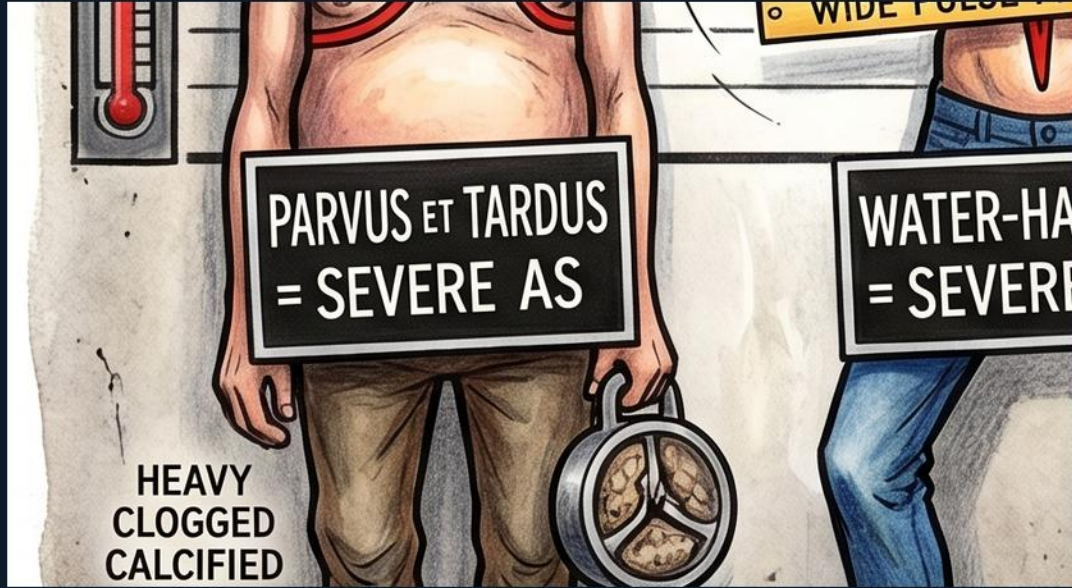


Cardiac Exam — Arterial Pulse Lineup





Parvus et tardus = severe AS

WHAT YOU SEE

The leftmost slumped old man with a flat low pulse tracing and placard 'PARVUS ET TARDUS = SEVERE AS'.

WHAT IT ENCODES

Pulsus parvus et tardus is a SLOW-RISING (tardus), LOW-AMPLITUDE (parvus) carotid pulse — the hallmark of SEVERE AORTIC STENOSIS. The stenotic valve blunts and delays ejection, weakening and slowing the upstroke.

THE BOARD TRAP

Parvus et tardus is AORTIC STENOSIS, NOT aortic regurgitation — AR gives the opposite (sharp, bounding, collapsing) pulse.



Slow low pulse tracing

WHAT YOU SEE

The faint, rounded, low-amplitude red waveform across the first man's chest.

WHAT IT ENCODES

The blunted upstroke and reduced peak reflect the fixed obstruction of AS — flow is throttled across the narrowed valve, so the arterial pressure rises slowly and never reaches a normal peak.

THE BOARD TRAP

A weak, delayed upstroke is obstruction (AS); don't read low amplitude alone as hypovolemia without the slow rise.



Heavy calcified valve

WHAT YOU SEE

A clogged, calcified valve icon at the first man's feet labeled 'HEAVY CLOGGED CALCIFIED VALVE'.

WHAT IT ENCODES

Severe calcific AORTIC STENOSIS is the lesion behind parvus et tardus — a heavily calcified, narrowed valve raises the resistance to LV ejection and produces the slow, small pulse.

THE BOARD TRAP

The calcified obstructing valve is the cause of the weak slow pulse — the underlying valve, not a vascular tone problem.



NOT AR

WHAT YOU SEE

A red 'X' over the words 'NOT AR' below the first man.

WHAT IT ENCODES

This explicitly rules out aortic regurgitation for the parvus-et-tardus pulse. AS = weak/slow; AR = bounding/collapsing — the two are bedside opposites and a classic distractor pairing.

THE BOARD TRAP

The tested wrong answer is AR — a slow, weak, delayed pulse is AS, never the bounding water-hammer of AR.



Water-hammer = severe AR

WHAT YOU SEE

The wild-eyed man swinging a mallet, with a sharp spiking waveform and placard 'WATER-HAMMER = SEVERE AR'.

WHAT IT ENCODES

The water-hammer (Corrigan) pulse has a SHARP, brisk RISE and a RAPID COLLAPSE — the classic finding of SEVERE AORTIC REGURGITATION. Regurgitant runoff into the LV in diastole drops aortic diastolic pressure abruptly.

THE BOARD TRAP

A bounding, collapsing pulse is AR, NOT AS — pairing it with AS is the trap; AS is slow and weak.



Mallet / hammer blow

WHAT YOU SEE

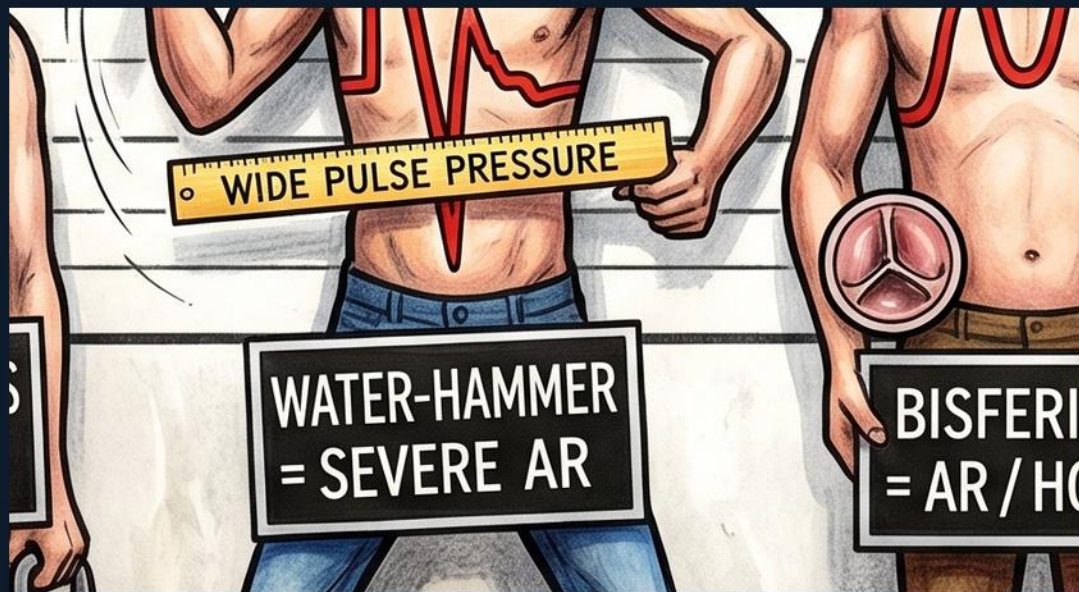
The raised mallet about to strike, depicting the percussive 'hammer' character of the pulse.

WHAT IT ENCODES

The hammer blow encodes the abrupt, forceful, percussive upstroke of the Corrigan pulse — a sudden hit followed by an equally sudden collapse, felt strikingly at the wrist with the arm raised.

THE BOARD TRAP

The forceful collapsing quality is AR runoff, not a hyperdynamic AS pulse — AS cannot produce a collapsing pulse.



Wide pulse pressure

WHAT YOU SEE

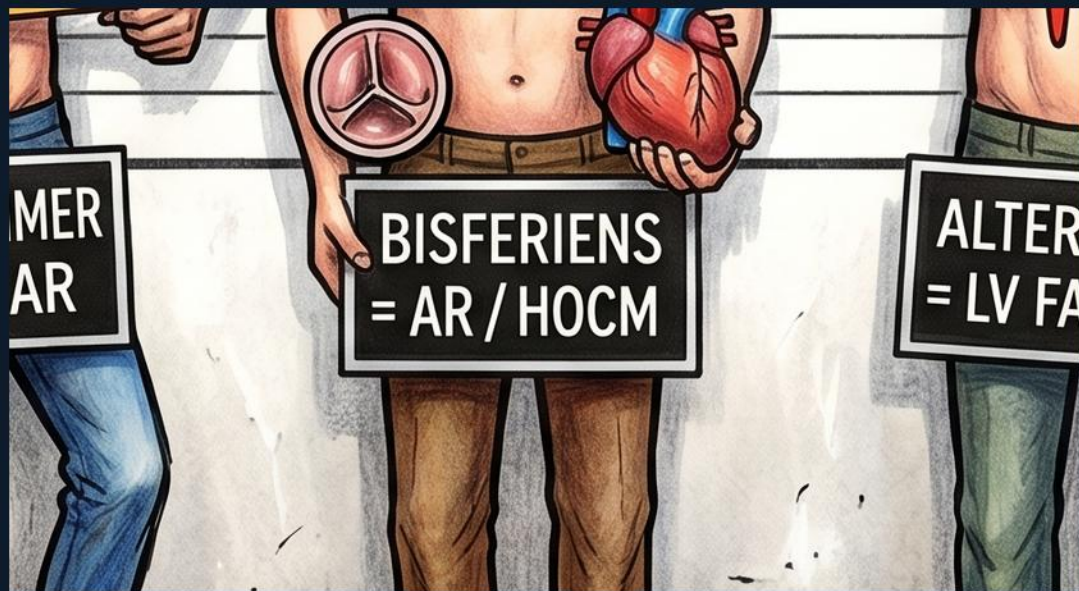
A yellow banner across the second man reading 'WIDE PULSE PRESSURE'.

WHAT IT ENCODES

Severe AORTIC REGURGITATION causes a WIDE PULSE PRESSURE — high systolic and low diastolic pressure — driving the peripheral signs (water-hammer pulse, Quincke's, de Musset's, Traube's).

THE BOARD TRAP

Wide pulse pressure goes with AR; severe AS narrows the pulse pressure — don't swap them.



Bisferiens = AR / HOCM

WHAT YOU SEE

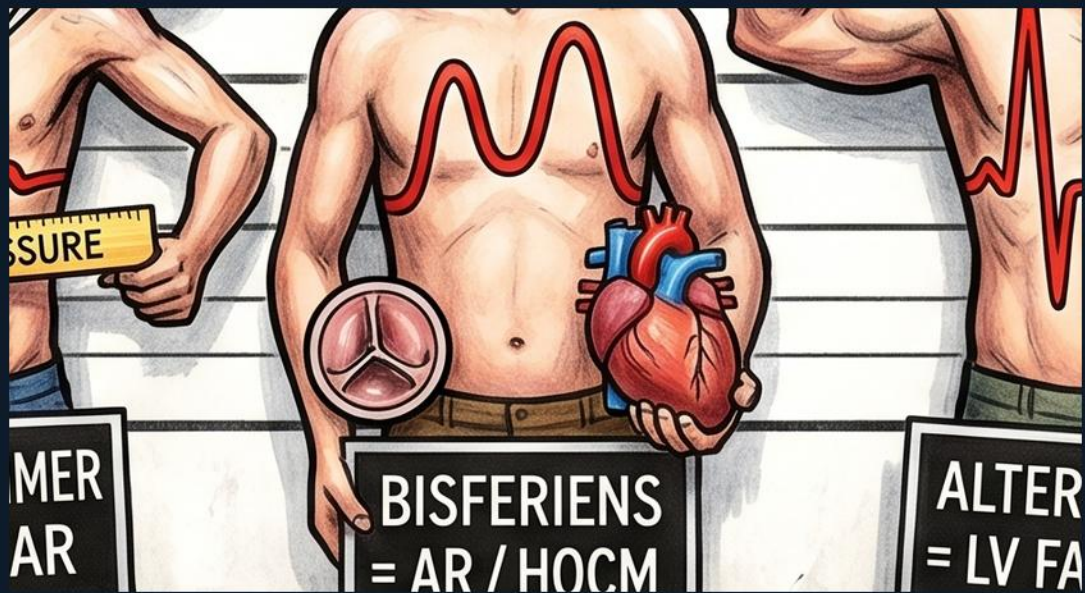
The center man with a TWO-PEAKED systolic waveform and placard 'BISFERIENS = AR / HOCM'.

WHAT IT ENCODES

Pulsus bisferiens has TWO SYSTOLIC PEAKS in one pulse. It is seen in AORTIC REGURGITATION, HOCM (dynamic outflow obstruction), or MIXED AS + AR. The mid-systolic dip separates the two percussion/tidal peaks.

THE BOARD TRAP

Bisferiens is NOT pure aortic stenosis — it points to AR, HOCM, or mixed AS+AR; pure AS gives parvus et tardus.



Two systolic peaks tracing

WHAT YOU SEE

The double-humped red waveform across the center man's chest.

WHAT IT ENCODES

The two peaks reflect a percussion wave then a tidal wave within a single systole — produced by the rapid ejection of AR/mixed disease or the mid-systolic obstruction dynamics of HOCM.

THE BOARD TRAP

Two peaks $\neq$ two beats — bisferiens is one pulse with twin peaks, not an arrhythmia or pulsus alternans.



NOT AS only

WHAT YOU SEE

A red 'X' over 'NOT AS ONLY' below the center man.

WHAT IT ENCODES

This rules out isolated aortic stenosis as the cause of bisferiens. Pure AS produces a single slow small peak; twin peaks require AR, HOCM, or combined AS+AR physiology.

THE BOARD TRAP

The trap answer is 'AS only' — bisferiens specifically implies more than isolated AS.



Alternans = LV failure

WHAT YOU SEE

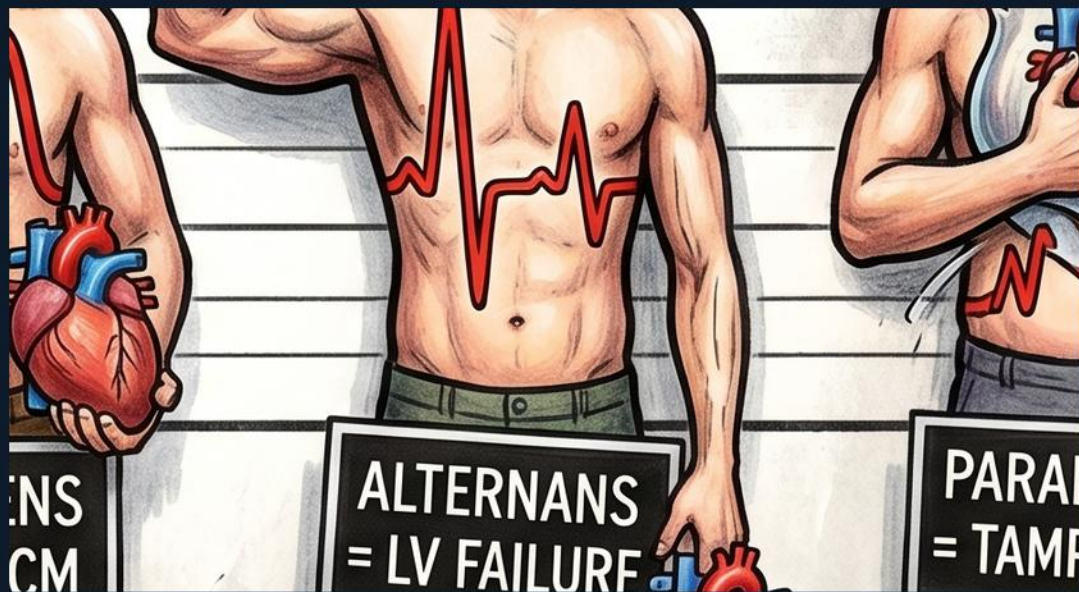
The flexing man with a tall-then-short alternating waveform and placard 'ALTERNANS = LV FAILURE'.

WHAT IT ENCODES

Pulsus alternans is a regular rhythm with beats of ALTERNATING STRONG and WEAK amplitude. It signals SEVERE LEFT VENTRICULAR SYSTOLIC FAILURE — the failing LV cannot generate equal stroke volume each beat.

THE BOARD TRAP

Pulsus alternans is LV systolic failure, NOT an arrhythmia — the rhythm is regular; only the amplitude alternates.



Strong/weak alternating beats

WHAT YOU SEE

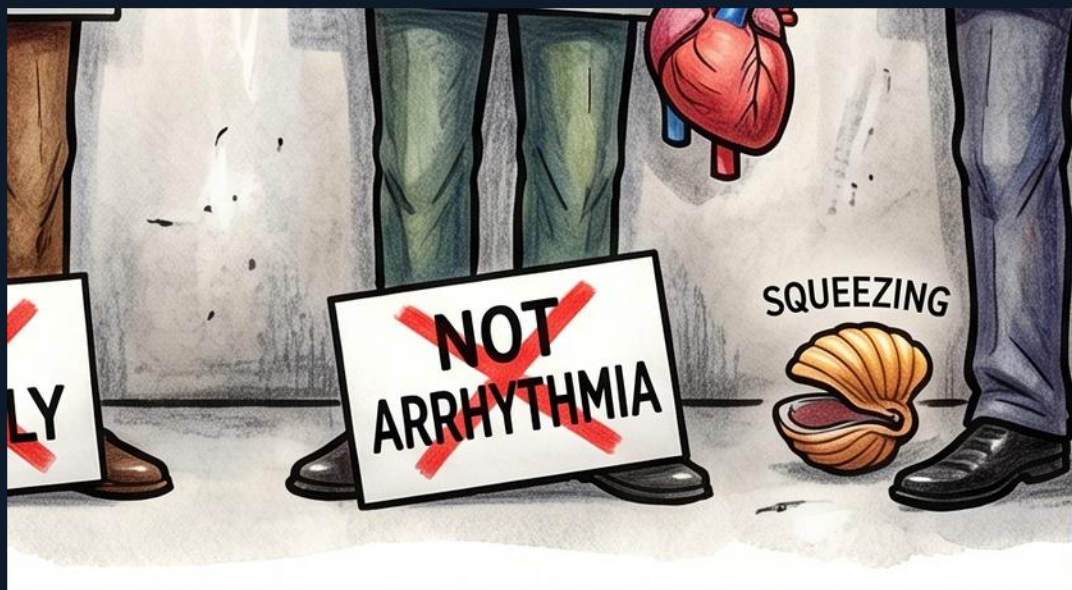
The alternating tall and short spikes of the fourth man's tracing.

WHAT IT ENCODES

The big-small-big-small amplitude pattern at a constant rate reflects beat-to-beat variation in LV stroke volume from a severely depressed, failing ventricle.

THE BOARD TRAP

Don't call it bigeminy — the timing is regular, so this is alternans (pump failure), not an ectopic rhythm.



NOT arrhythmia

WHAT YOU SEE

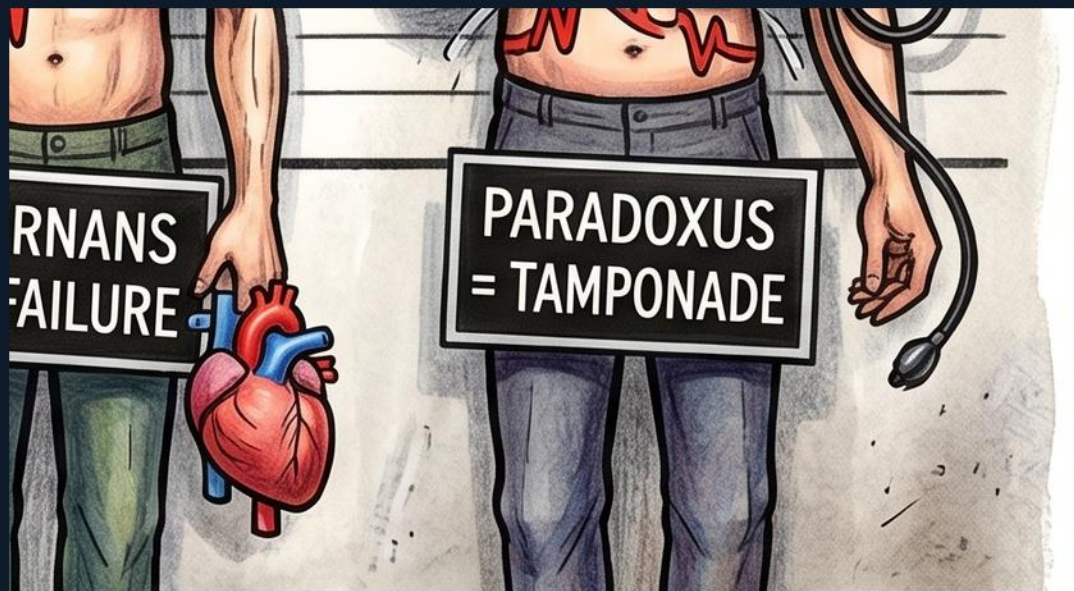
A red 'X' over 'NOT ARRHYTHMIA' below the fourth man.

WHAT IT ENCODES

This rules out an arrhythmic explanation for alternans. The pulse alternates in strength at a regular rate — the problem is contractile (severe LV systolic dysfunction), not electrical.

THE BOARD TRAP

The trap answer is 'arrhythmia' — alternans has a normal regular rhythm; only amplitude varies.



Paradoxus = tamponade

WHAT YOU SEE

The rightmost man clutching his chest in a windstorm with placard 'PARADOXUS = TAMPONADE' and a BP cuff.

WHAT IT ENCODES

Pulsus paradoxus is an EXAGGERATED FALL in systolic BP (>10 mmHg) during INSPIRATION. Classic in CARDIAC TAMPONADE; also constrictive pericarditis, severe ASTHMA, and COPD. Inspiratory RV filling bows the septum and limits LV stroke volume.

THE BOARD TRAP

It is a quantified inspiratory DROP (>10 mmHg), not just any respiratory variation — and it is exaggeration of a normal phenomenon, not a true 'paradox'.



Inspiration drops SBP

WHAT YOU SEE

Wind labeled 'INSPIRATION' and 'INHALE' blowing across the last man with a sphygmomanometer.

WHAT IT ENCODES

On inspiration, increased venous return fills the RV, the interventricular septum shifts left in the constrained pericardial space, LV filling and stroke volume fall, and systolic BP drops more than 10 mmHg — measured with the cuff.

THE BOARD TRAP

Pulsus paradoxus is detected by BP cuff (>10 mmHg inspiratory fall), not by palpation alone — and tamponade is the headline cause.

Summary — every symbol

1. Parvus et tardus = severe AS

Pulsus parvus et tardus is a SLOW-RISING (tardus), LOW-AMPLITUDE (parvus) carotid pulse — the hallmark of SEVERE AORTIC STENOSIS

3. Heavy calcified valve

Severe calcific AORTIC STENOSIS is the lesion behind parvus et tardus — a heavily calcified, narrowed valve raises the resistance to LV ejection and produces the slow, small pulse.

5. Water-hammer = severe AR

The water-hammer (Corrigan) pulse has a SHARP, brisk RISE and a RAPID COLLAPSE — the classic finding of SEVERE AORTIC REGURGITATION

7. Wide pulse pressure

Severe AORTIC REGURGITATION causes a WIDE PULSE PRESSURE — high systolic and low diastolic pressure — driving the peripheral signs (water-hammer pulse, Quincke's, de Musset's, Traube's).

9. Two systolic peaks tracing

The two peaks reflect a percussion wave then a tidal wave within a single systole — produced by the rapid ejection of AR/mixed disease or the mid-systolic obstruction dynamics of HOCM.

11. Alternans = LV failure

Pulsus alternans is a regular rhythm with beats of ALTERNATING STRONG and WEAK amplitude

13. NOT arrhythmia

This rules out an arrhythmic explanation for alternans

15. Inspiration drops SBP

On inspiration, increased venous return fills the RV, the interventricular septum shifts left in the constrained pericardial space, LV filling and stroke volume fall, and systolic BP drops more than 10 mmHg — measured with the cuff

2. Slow low pulse tracing

The blunted upstroke and reduced peak reflect the fixed obstruction of AS — flow is throttled across the narrowed valve, so the arterial pressure rises slowly and never reaches a normal peak.

4. NOT AR

This explicitly rules out aortic regurgitation for the parvus-et-tardus pulse

6. Mallet / hammer blow

The hammer blow encodes the abrupt, forceful, percussive upstroke of the Corrigan pulse — a sudden hit followed by an equally sudden collapse, felt strikingly at the wrist with the arm raised.

8. Bisferiens = AR / HOCM

Pulsus bisferiens has TWO SYSTOLIC PEAKS in one pulse

10. NOT AS only

This rules out isolated aortic stenosis as the cause of bisferiens

12. Strong/weak alternating beats

The big-small-big-small amplitude pattern at a constant rate reflects beat-to-beat variation in LV stroke volume from a severely depressed, failing ventricle.

14. Paradoxus = tamponade

Pulsus paradoxus is an EXAGGERATED FALL in systolic BP (>10 mmHg) during INSPIRATION